「Evaluate

$$\int_{-\infty}^{\infty} e^{-\pi x^2}\cdot e^{-2\pi i x\xi}\,dx = e^{-\pi\xi^2}. \qquad (\xi>0)$$

Proof. Let $f(z)=e^{-\pi z^2}$. Then, it is entire.

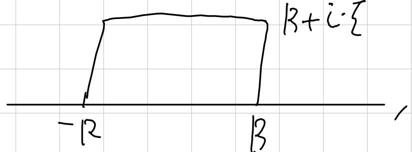
Consider four contour: $C_1: t \ (-R\le t\le R)$,

$C_2: R+it \ (0\le t\le \xi)$

$C_3: t+i\xi \ (-R\le t\le R)$

$C_4: -R+it \ (0\le t\le \xi)$

Then, the Composition $C=C_1\cdot C_2\cdot(-C_3)\cdot(-C_4)$ is described as



and, by the Cauchy's theorem, $\int_C f=0$. Now,

$$\int_{C_1} f = \int_{-R}^{R} e^{-\pi x^2}\,dx,$$

$$\int_{C_2} f = \int_0^{\xi} e^{-\pi(R+it)^2}\cdot i\,dt = i\cdot\int_0^{\xi} e^{-\pi(R^2-t^2+2iRt)}\cdot dt$$

and so $\left|\int_{C_2} f\right| \fallingdotseq \left|\int_0^{\xi} e^{\pi\cdot(t^2-R^2)}\,dt\right| \le \left|\int_0^{\xi} e^{\pi t^2}\,dt\right|\cdot e^{-\pi R^2}\to 0$ as $R\to\infty$.

Similarly $\left|\int_{C_4} f\right|\to 0$ as $R\to\infty$. And,

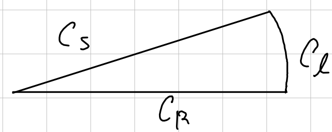
$$\int_{C_3} f = \int_{-R}^{R} e^{-\pi(t+i\xi)^2}\,dt = \int_{-R}^{R} e^{-\pi(t^2-\xi^2+2it\xi)}\,dt$$

$$= e^{\pi\xi^2}\cdot\int_{-R}^{R} e^{-\pi t^2}\cdot e^{-2\pi i t\xi}\,dt$$

Thus, as $R\to\infty$,

$$0=\int_{-\infty}^{\infty} e^{-\pi x^2}\,dx - e^{\pi\xi^2}\int_{-\infty}^{\infty} e^{-\pi t^2}\cdot e^{-2\pi i t\xi}\,dt. \qquad \boxtimes$$

「For each $n=2,3 \cdots$ evaluate
$$\int_0^\infty \frac{1}{1+x^n}\,dx.$$



proof. Let $f(z)=\frac{1}{1+z^n}$. Take a contour C as a composition of:

$C_R: t\mapsto t\ (0\le t\le R)$, $C_\ell: t\mapsto R\cdot e^{\frac{it}{n}}\ (0\le t\le 2\pi)$,

$C_s: t\mapsto t\cdot e^{\frac{2\pi i}{n}}\ (0\le t\le R)$, that is, $C = C_R * C_\ell * (-C_s)$.

Inside C, $1+z^n$ has only one zero, $\exp(i\pi/n)$, so f has only one singularity inside C, the simple pole $\exp(i\pi/n)$, by the Residue theorem,
$$\int_C f = 2\pi i\cdot \mathrm{Res}(f,\exp(i\pi/n)) = \lim_{z\to\exp(i\pi/n)} (z-\exp(i\pi/n))\cdot\frac{1}{1+z^n}\cdot 2\pi i$$
$$= \frac{2\pi i}{n\cdot\exp(i\pi/n)^{n-1}} = -\frac{\exp(i\pi/n)}{n}\cdot 2\pi i$$

Meanwhile,
$$\left|\int_{C_\ell} f\right| = \int_0^{2\pi}\left|\frac{1}{1+R^n\cdot e^{it}}\frac{Ri}{n}\cdot e^{\frac{it}{n}}\right|dt$$
$$= \int_0^{2\pi}\frac{R/n}{|1+R^n\cdot e^{it}|}\,dt \le \int_0^{2\pi}\frac{R/n}{|R^n-1|}\,dt \to 0 \text{ as } R\to\infty.$$

and
$$\int_{C_s} f = \int_0^R \frac{e^{2\pi i/n}}{1+t^n e^{2\pi i}}\,dt = e^{\frac{2\pi i}{n}}\cdot\int_0^R\frac{1}{1+t^n}\,dt.$$
So that
$$-\frac{2\pi i}{n}\cdot\exp(i\pi/n) = \lim_{R\to\infty}\int_C f = \int_0^\infty \frac{1}{1+x^n}\,dx - \int_0^\infty\frac{1}{1+x^n}\,dx\cdot e^{\frac{2\pi i}{n}}$$
$$= \left(1-e^{\frac{2\pi i}{n}}\right)\cdot\int_0^\infty\frac{1}{1+x^n}\,dx$$
$$\Rightarrow \int_0^\infty\frac{1}{1+x^n}\,dx = -\frac{2\pi i}{n}\cdot\exp(i\pi/n)\Big/\left(1-e^{\frac{2\pi i}{n}}\right).$$

「Evaluate $\int_0^\infty \frac{x^2}{1+x^6}\,dx$

Proof. Let C_R be a closed curve of composition $C_R = C_x * C_r * (-C_s)$ where
$C_x: t\mapsto t$ $(0\le t\le R)$, $C_r: t\mapsto R\cdot e^{i\pi t/6}$ $(0\le t\le 2\pi)$, $C_s: t\mapsto te^{\pi i/3}$ $(0\le t\le R)$
Inside the curve C, $f(z)=z^2/(1+z^6)$ has only one singularity at $z_0=\exp(\pi i/6)$. By the Residue theorem

$$\int_{C_R} f = 2\pi i\cdot \mathrm{Res}(f,z_0) = 2\pi i\cdot \lim_{z\to z_0}(z-z_0)\cdot\frac{z^2}{1+z^6} = z_0^2\cdot\frac{2\pi i}{6z_0^5} = \frac{2\pi i}{6z_0^3}=\frac{\pi}{3}$$

Meanwhile, for $R>1$,

$$\left|\int_{C_r} f\right| \le \int_0^{2\pi}\left|\frac{R^2 e^{i\pi t/3}}{1+R^6\cdot e^{i\pi t}}\right|\left|\frac{\pi i}{6}\cdot R\cdot e^{i\pi t/6}\right|dt$$

$$\le \int_0^{2\pi}\frac{R^3}{|R^6-1|}\cdot\frac{\pi}{6}\,dt \to 0 \text{ as } R\to\infty$$

$$\int_{C_s} f = \int_0^R \frac{x^2\cdot e^{2\pi i/3}}{1+x^6}\cdot e^{\pi i/3}\cdot dt = -\int_0^R\frac{x^2}{1+x^6}\,dx$$

Consequently,

$$\frac{1}{3\pi} = \lim_{R\to\infty}\int_{C_R} f = 2\cdot\int_0^\infty \frac{x^2}{1+x^6}\,dx,$$ so $\frac{\pi}{6}$ is we desired. $\square$

Evaluate $\int_0^\infty \sin(x^2)\,dx$. Is it equals to $\int_0^\infty \cos(x^2)\,dx$

Evaluate $\int_0^\infty \frac{\sin x}{x}\,dx$, using indented semicircle.

Evaluate $\int_0^\infty \frac{\log x}{1+x^4}\,dx$